BetaNav | Frozen VLM + Prefix-KV Flow Action Expert

Trained architecture / 30 future actions / Beta(1, 1.5) / 1.0 × FM loss

ACTION EXPERT 60.86M

01 FROZEN CONDITION ENCODER

Instruction + image history through t
Slow–Fast frame sampling / spatial pooling

Original LightNav / Qwen3-VL
Vision encoder + 36 language layers

eval() · no_grad() · 4.448B frozen parameters

VLM layer 6 → native K / V
VLM layer 12 → native K / V
VLM layer 18 → native K / V
VLM layer 24 → native K / V
VLM layer 30 → native K / V
VLM layer 36 → native K / V

Each selected K and V: [B, 8, N, 128]
Native K normalization + multimodal RoPE
N = padded visual / language prefix length

Capture only the 6 selected KV pairs

detach

02 TRAINABLE ACTION EXPERT

Noisy actions x_τ : [B, 30, 3]

Linear 3 → 512 + learned step embedding
Hidden actions: [B, 30, 512]

6 blocks · residual width 512 · SwiGLU width 2048

Action block 1 · KV from layer 6
Action block 2 · KV from layer 12
Action block 3 · KV from layer 18
Action block 4 · KV from layer 24
Action block 5 · KV from layer 30
Action block 6 · KV from layer 36

Final time AdaRMSNorm → Linear 512 → 3

Flow field $v_\theta(x_\tau, \tau, C)$: [B, 30, 3]

03 INSIDE EACH ACTION BLOCK

Time τ → sinusoidal embedding (512)
→ shared MLP 512 → 512 → 512

Time controls scale, shift and residual gates

Time AdaRMSNorm (attention)

Action Q: 32 × 128 = 4096
Action K / V: 8 × 128 = 1024
Q / K normalization + action RoPE

ONE joint attention / ONE softmax
Q = action queries
K / V = [frozen prefix ; noisy actions]

All 30 actions attend bidirectionally

Output 4096 → 512 · gated residual

Time AdaRMSNorm (FFN)

SwiGLU 512 → 2048 → 512
+ gated residual

9,964,800 parameters per block

TRAINING / FM loss updates only the orange expert

Normalized GT a: 30 local SE(2) increments; $\varepsilon \sim \mathcal{N}(0, I)$

$$\tau \sim \mathrm{Beta}(1, 1.5), \qquad x_\tau = (1-\tau)\varepsilon + \tau a$$

$$\mathcal{L}_{\mathrm{FM}} = \mathrm{MSE}_{\mathrm{valid}}(v_\theta,\ a - \varepsilon)$$

τ = 0: noise / τ = 1: clean actions
Mask invalid GT tail steps in the loss. No CE or RVQ targets.

INFERENCE / reuse the same prefix KV

Encode the current instruction / history once → cache 6 KV pairs

$$x_0 \sim \mathcal{N}(0, I), \quad x_{k+1} = x_k + \tfrac{1}{10} v_\theta(x_k, k/10, C)$$

10 Euler iterations → denormalize → 30 × 3 local increments

Compose increments into waypoints.
30 future action steps and 10 flow iterations are independent.

Trainable: 60,858,371 parameters (1.35% of the full 4.509B model)

Source: completed step 3699 metadata + current prefix_kv.py / pi_action_expert.py / beta_model.py

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